Avishek Das

Ph.D. Candidate in Computer Science

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OVERVIEW

36	8	7	6
publications	first-author papers	grant proposals co-written	journal reviews

EDUCATION

Kansas State University, Manhattan, KS, USA — Jul 2024 – Present
Ph.D. in Computer Science (Ph.D. Candidate) — GPA: 4.00/4.00
Advisor: Dr. Pascal Hitzler, Data Semantics Lab (DaSe Lab) — Expected: Summer 2028

Chittagong University of Engineering and Technology — Jan 2023 – Jun 2025
M.Sc. in Computer Science & Engineering — CGPA: 3.62/4.00
Chattogram, Bangladesh

Chittagong University of Engineering and Technology — Feb 2016 – Jun 2021
B.Sc. in Computer Science & Engineering — CGPA: 3.80/4.00 (with Honors)
Chattogram, Bangladesh; class rank: 4th of 120

RESEARCH INTERESTS

Explainable and Neurosymbolic AI; Model and LLM Interpretability; Knowledge Graphs; LLM Agents and Agentic AI; Computer Vision and Natural Language Processing; MLOps.

RESEARCH EXPERIENCE

Graduate Research Assistant, Kansas State University — Jul 2024 – Present
Data Semantics Lab (DaSe Lab); Supervisor: Dr. Pascal Hitzler — Manhattan, KS, USA

- Develop neurosymbolic interpretability methods that label what individual hidden neurons in deep networks detect, using concept induction (DL-Learner) over OWL knowledge graphs to turn neuron activations into human-readable concepts.
- Extended concept-induction neuron analysis from image models to text classifiers (first-author paper at K-CAP 2025) and released the text_hna toolkit.
- Analyze how CNNs represent contextual image patches in agricultural datasets to expose context bias, combining OWL knowledge graphs with DL-Learner concept induction (preprint, 2026).
- Investigating the internal representations of open-weight LLMs (Gemma) by combining sparse autoencoders with concept induction to label learned features with human-readable concepts (ongoing).
- Project: *Mapping Seed-to-Plant Life Cycle to Predict Yields – Integrated Artificial Intelligence and Data Analytics Approach* (funded by the Kansas State University Game-changing Research Initiation Program, GRIP).

Research Coordinator, CUET NLP Lab — Jun 2021 – Jun 2024
Chittagong University of Engineering and Technology — Chattogram, Bangladesh

- Supervised 2 and co-supervised 25 undergraduate researchers in NLP, multimodal learning and LLMs, producing about 20 peer-reviewed papers at venues including EACL, EMNLP workshops and ICCIT.
- Fine-tuned transformer models (BERT, MuRIL, multilingual masked language models) in PyTorch and Hugging Face for emotion, sentiment, hate-speech and fake-news detection in low-resource languages (Bengali, Tamil, Telugu, Malayalam).

PROPOSAL WRITING EXPERIENCE

Grant proposals (co-writer)

HARVESTGraph: A Multimodal Knowledge Graph for AI-Enabled Agricultural Discovery
National Science Foundation (NSF) — Submitted 11/06/2026

Beyond Robustness Testing: Neurosymbolic Explanation and Suppression of Deep Neural Network Failures
Amazon — Submitted 11/06/2026

E-RISE: Advancing Resilience in Kansas (ARK): Building AI-Assisted Microbial Solutions for Water and Soil Systems
National Science Foundation (NSF) — Submitted 08/12/2026

Physics-informed AI for Quantum Interactions (PAIQ)
U.S. Department of Energy — Submitted 06/10/2026

FAIR Science Data for the Proto-OKN (FASeD)
National Science Foundation (NSF) — Submitted 04/08/2026

Adaptive Management to Strengthen Agricultural Systems
USDA Research, Education, and Economics – National Institute of Food and Agriculture (NIFA); Solicitation USDA-NIFA-AFRI-011677 — Submitted 03/16/2026

PCL-Test Bed: CAVES-AI: Collaborative Agricultural Virtual Experiment Station – A Programmable Cloud Laboratory for AI-Driven Breeding and Just-in-Time Data Synthesis
National Science Foundation (NSF) — Submitted 11/21/2025

Non-grant proposals

Understanding What Neural Networks Learn: Semantic Alignment of Latent Representation for Explainable AI
Thematic Special Session proposal, XAI World Conference 2027

TEACHING EXPERIENCE

Lecturer, Department of Computer Science & Engineering, CUET — Jul 2022 – Jun 2024
Chittagong University of Engineering and Technology — Chattogram, Bangladesh
Taught and co-taught theory and laboratory courses in *Artificial Intelligence*, *Computer Programming and Engineering Analysis*, *Compiler Design* and *Software Engineering* to about 125 students per term.

AI Consultant and Python Instructor, Diligite Ltd. — May 2021 – May 2022
Remote — Bangladesh
Guided ML-based projects and taught Python fundamentals for machine learning.

INDUSTRY EXPERIENCE

Assessment Developer, Workera — Sep 2023 – Jun 2024
Remote — Palo Alto, CA, USA

- Designed about 20 skill assessments and co-developed about 25 more (about 100 items each) as primary or secondary subject matter expert in NLP, Computer Vision, MLOps, Responsible AI and Cybersecurity.
- Incorporated large language models into the assessment-authoring workflow to automate item drafting.
- Contributed to score calibration and the standardization of learning pathways for computer adaptive tests.

Machine Learning Engineer, Apurba Technologies Ltd. — Jul 2021 – Jul 2022
Dhaka, Bangladesh

- Built a Bangla OCR system with document layout reconstruction, deployed in the Government of Bangladesh's national OCR service (ocr.bangla.gov.bd) under the EBLICT project of the Bangladesh Computer Council, ICT Division.
- Co-designed an attention-based multi-headed CNN for grapheme-level recognition trained on 1.37M character images, reaching 99.6% accuracy on computer-composed, 99.3% on letterpress, 98.0% on typewritten and 95.3% on handwritten text (IEEE Big Data 2021).
- Built a multi-headed ResUNet++ recognizer that generalizes across scripts: 95.4% on Bangla, 94.7% on Assamese and 96.4% on English (EMNIST), outperforming prior published results (ICPR 2022, first author).
- Optimized and deployed models to production and processed large-scale document image data.

Trainee Software Developer, Brain Station 23 Ltd. Aug 2019
Dhaka, Bangladesh

Designed and implemented a University Enrollment Log System for student–course assignment (Django, AJAX, D3.js, JavaScript).

PUBLICATIONS

P = Preprint, B = Book Chapter, J = Journal, C = Conference, W = Workshop; * = Avishek Das

Preprints

[P.2] A. Dalal, **A. Das***, S. E. Akkamahadevi, "Understanding Contextual Bias in Agricultural CNNs Through Hidden-Neuron Concept Induction," *arXiv preprint, 2026*.

[P.1] **A. Das***, P. Hitzler, "Bridging Neural Representations and Symbolic Semantics in Audio Classification," *arXiv:2610.40414 [cs.SD], 2026*.

Book Chapters

[B.2] A. Barua, **A. Das***, M. S. Sarma, P. Hitzler, "Reasoning in Large Language Models: RAG and Beyond," *Neuro-Symbolic AI: Bridging the Gap Between Neural Networks and Symbolic Reasoning, Elsevier, 2026*.

[B.1] A. Dalal, M. S. Sarma, **A. Das***, S. E. Akkamahadevi, E. Y. Vasserman, P. Hitzler, "Neurosymbolic Hidden Neuron Analysis in Convolutional Neural Networks," *Neuro-Symbolic AI: Bridging the Gap Between Neural Networks and Symbolic Reasoning, Elsevier, 2026*.

Journal Articles

[J.4] A. Dalal, R. Rayan, A. Barua, S. E. Akkamahadevi, **A. Das***, C. Widmer, E. Y. Vasserman, M. K. Sarker, P. Hitzler, "Toward a Neurosymbolic Understanding of Hidden Neuron Activations," *Neurosymbolic Artificial Intelligence, vol. 2, 2026*.

[J.3] **A. Das***, M. S. Sarma, M. M. Hoque, N. Siddique, M. Dewan, "AVaTER: Fusing Audio, Visual, and Textual Modalities Using Cross-Modal Attention for Emotion Recognition," *Sensors, vol. 24, no. 18, p. 5862, 2024*.

[J.2] **A. Das***, M. M. Hoque, O. Sharif, M. A. Akbar, N. Siddique, "TEmoX: Classification of Textual Emotion based on Ensemble of Transformers," *IEEE Access*.

[J.1] M. A. Iqbal, **A. Das***, O. Sharif, M. M. Hoque, I. H. Sarker, "BEmoC: A Corpus for Identifying Emotion in Bengali Texts," *SN Computer Science, vol. 3, 2022*.

Conference Papers

[C.21] **A. Das***, A. Dalal, P. Hitzler, "Hidden Neuron Activation Analysis on Labeled Text Data," *13th Knowledge Capture Conference (K-CAP '25), 2025*.

[C.20] **A. Das***, M. S. Sarma, M. M. Hoque, "Multimodal Emotion Recognition Exploiting Audio, Video and Textual Features with Crossmodal Attention," *IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE), 2024*.

[C.19] S. Ahsan, E. Hossain, O. Sharif, **A. Das***, M. M. Hoque, M. Dewan, "A Multimodal Framework to Detect Target Aware Aggression in Memes," *18th Conference of the European Chapter of the Association for Computational Linguistics (EACL), 2024*.

[C.18] S. Ahsan, F. Tasnia, N. Tabassum, **A. Das***, M. M. Hoque, N. Siddique, "Classifying Textual Sentiment Using Bidirectional Encoder Representations from Transformers," *26th International Conference on Computer and Information Technology (ICCIT), 2023*.

[C.17] J. Hossain, **A. Das***, M. M. Hoque, N. Siddique, "Multilabel Aggressive Text Classification from Social Media using Transformer-based Approaches," *26th International Conference on Computer and Information Technology (ICCIT), 2023*.

[C.16] F. S. Tamim, S. Saha, **A. Das***, M. M. Hoque, "Detecting Violence Inciting Texts based on Pre-trained Transformers," *26th International Conference on Computer and Information Technology (ICCIT), 2023*.

[C.15] S. A. M. Hayat, **A. Das***, M. M. Hoque, "Abstractive Bengali Text Summarization

Using Transformer-based Learning," *26th International Conference on Computer and Information Technology (ICCIT), 2023*.

[C.14] S. S. Rahman, **A. Das**[*], O. Sharif, M. M. Hoque, "Identification of Deceptive Clickbait YouTube Videos Using Multimodal Features," *International Conference on Intelligent Computing & Optimization (ICO), 2023*.

[C.13] M. D. S. Rahman, O. Sharif, **A. Das**[*], S. Afroze, M. M. Hoque, "FaND-X: Fake News Detection using Transformer-based Multilingual Masked Language Model," *IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE)*.

[C.12] N. Jannat, **A. Das**[*], O. Sharif, M. M. Hoque, "DeepSen: A Deep Learning-based Framework for Sentiment Analysis from Multi-Domain Heterogeneous Data," *25th International Conference on Computer and Information Technology (ICCIT)*.

[C.11] N. Jannat, **A. Das**[*], O. Sharif, M. M. Hoque, "An Empirical Framework for Identifying Sentiment from Multimodal Memes using Fusion Approach," *25th International Conference on Computer and Information Technology (ICCIT)*.

[C.10] **A. Das**[*], A. S. A. Rabby, I. Kowsar, F. Rahman, "A Deep Learning Based Unified Solution for Character Recognition," *26th International Conference on Pattern Recognition (ICPR), 2022*.

[C.9] M. M. Islam, **A. Das**[*], I. Kowsar, A. S. A. Rabby, N. Hasan, F. Rahman, "Towards Building a Bangla Text Recognition Solution with a Multi-Headed CNN Architecture," *IEEE International Conference on Big Data, 2021*.

[C.8] A. Mahbub Uddin, A. A. Miraj, M. Sen Sarma, **A. Das**[*], M. M. Gani, "Traditional Bengali Food Classification Using Convolutional Neural Network," *IEEE Region 10 Symposium (TENSYMP)*.

[C.7] **A. Das**[*], O. Sharif, M. M. Hoque, I. H. Sarker, "Emotion Classification in a Resource Constrained Language Using Transformer-based Approach," *Conference of the North American Chapter of the Association for Computational Linguistics (NAACL): Student Research Workshop, 2021*.

[C.6] M. Kowsher, **A. Das**[*], M. H. Sarker, A. Tahabilder, M. Z. I. Sanjid, "SeqVectorizer: Sequence Representation in Vector Space," *4th International Conference on Networking, Information Systems & Security (NISS)*.

[C.5] M. Sen Sarma, **A. Das**[*], "BMGC: A Deep Learning Approach to Classify Bangla Music Genres," *4th International Conference on Networking, Information Systems & Security (NISS), ACM*.

[C.4] M. Kowsher, M. Z. Islam Sanjid, F. Afrin, **A. Das**[*], P. Saha, "Gender Identification from Bangla Name Using Machine Learning and Deep Learning Algorithms," *Emerging Technologies in Data Mining and Information Security, 2021*.

[C.3] **A. Das**[*], M. A. Iqbal, O. Sharif, M. M. Hoque, "BEmoD: Development of Bengali Emotion Dataset for Classifying Expressions of Emotion in Texts," *International Conference on Intelligent Computing & Optimization (ICO)*.

[C.2] F. Jahara, A. Barua, M. A. Iqbal, **A. Das**[*], O. Sharif, M. M. Hoque, I. H. Sarker, "Towards POS Tagging Methods for Bengali Language: A Comparative Analysis," *International Conference on Intelligent Computing & Optimization (ICO)*.

[C.1] M. Kowsher, M. Z. Islam Sanjid, **A. Das**[*], M. Ahmed, M. M. Hossain Sarker, "Machine Learning and Deep Learning-based Information Extraction from Bangla Names," *Procedia Computer Science, vol. 178, 9th International Young Scientists Conference in Computational Science*.

Workshop Papers

[W.7] A. Raihan, T. Rahman, M. Rahman, S. Ahsan, J. Hossain, **A. Das**[*], M. M. Hoque, "CUET_DUO@StressIdent_LT-EDI@EACL2024: Stress Identification Using Tamil-Telugu BERT," *Fourth Workshop on Speech, Vision, and Language Technologies for Dravidian Languages @ EACL 2024*.

[W.6] Z. Tripty, M. Nafis, A. Chowdhury, S. Ahsan, J. Hossain, **A. Das**[*], M. M. Hoque, "CUETSentimentSillies@DravidianLangTech-EACL2024: Transformer-based Approach for Sentiment Analysis in Tamil and Tulu Code-Mixed Texts," *Fourth Workshop on Speech, Vision, and Language Technologies for Dravidian Languages @ EACL 2024*.

[W.5] S. Farsi, A. Eusha, T. Rahman, S. Ahsan, J. Hossain, **A. Das**[*], M. M. Hoque, "CUET_Binary_Hackers@DravidianLangTech EACL2024: Hate and Offensive Language Detection in Telugu Code-Mixed Text Using Sentence Similarity BERT," *Fourth Workshop on Speech, Vision, and Language Technologies for Dravidian Languages @ EACL 2024*.

[W.4] T. Rahman, A. Raihan, M. Rahman, J. Hossain, S. Ahsan, **A. Das**[*], M. M. Hoque, "CUET_DUO@DravidianLangTech EACL2024: Fake News Classification Using Malayalam-BERT," *Fourth Workshop on Speech, Vision, and Language Technologies for Dravidian Languages @ EACL 2024*.

[W.3] S. Farsi, A. Eusha, A. Islam, H. M. A. Taher, J. Hossain, S. Ahsan, **A. Das**[*], M. M. Hoque, "CUET_Binary_Hackers@DravidianLangTech EACL2024: Fake News Detection in Malayalam Language Leveraging Fine-tuned MuRIL BERT," *Fourth Workshop on Speech, Vision, and Language Technologies for Dravidian Languages @ EACL 2024*.

[W.2] K. Ahmed, M. Osama, M. S. Islam, M. T. Islam, **A. Das**[*], M. M. Hoque, "NLP_CUET at BLP-2023 Task 1: Fine-grained Categorization of Violence Inciting Text using Transformer-based Approach," *First Workshop on Bangla Language Processing (BLP) @ EMNLP 2023*.

[W.1] J. Hossain, H. M. A. Taher, **A. Das**[*], M. M. Hoque, "Score_IsAll_You_Need at BLP-2023 Task 1: A Hierarchical Classification Approach to Detect Violence Inciting Text using Transformers," *First Workshop on Bangla Language Processing (BLP) @ EMNLP 2023*.

SELECTED PROJECTS

ML Knowledge Hub: Agentic Hybrid GraphRAG for ML Project Knowledge

- Agentic GraphRAG assistant combining a Neo4j knowledge graph with Qdrant semantic search to answer questions about ML projects, models, datasets and evaluations with cited sources.
- Planner–critic–revision agent pipeline turns each question into a structured query plan (96.7% routing accuracy on a 30-query benchmark); supports multi-turn chat. Deployed on Streamlit and exposed to AI clients through an MCP server. Live demo: mlknowledge.streamlit.app.
- Tools: Python, OpenAI API, Neo4j, Qdrant, Sentence-Transformers, Streamlit, MCP, Pydantic, pytest.

text_hna: Hidden Neuron Activation Analysis on Labeled Text Data
Concept-induction pipeline that explains how a deep model makes text-classification decisions (Python, TensorFlow).

TransEmoC: A Transformer-Based Unified Toolkit for Emotion Classification
Toolkit for training and predicting textual emotion, with pretrained weights for Bengali (BERT, TensorFlow).

IoT-Based Face Recognition Attendance System
Records attendance by recognizing students' faces from a webcam at the classroom entrance (Raspberry Pi, CNN, OpenCV, PyQt5).

REVIEWER ACTIVITIES

Conference: Subreviewer (Main Track), World Conference on eXplainable Artificial Intelligence (xAI 2026).
Journals: *IEEE Access* (5 reviews); *IEEE Transactions on Human-Machine Systems* (1 review).

TECHNICAL SKILLS

- **Programming:** Python, C, C++, Java, JavaScript, SQL, Bash.
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Hugging Face Transformers, fine-tuning transformer models (BERT, MuRIL), CNNs, multimodal learning (cross-modal

attention), embedding models, time-series forecasting (LSTM), multi-sensor fusion.
- **Interpretability:** Explainable AI (XAI), neurosymbolic AI, concept induction (DL-Learner), sparse autoencoders, hidden-neuron activation analysis.
- **NLP & Computer Vision:** text classification, sentiment and emotion analysis, summarization, OCR, document layout analysis.
- **Generative AI & LLMs:** agentic and multi-agent systems, RAG and GraphRAG, Model Context Protocol (MCP), LLM evaluation, structured outputs, OpenAI API.
- **Knowledge Graphs & Databases:** Neo4j (Cypher), GraphDB (Ontotext), RDF/OWL, SPARQL, ontology engineering, vector databases (Qdrant).
- **Engineering & Deployment:** Docker, Git, pytest, AWS (EC2, S3), GCP, HPC/SLURM, Streamlit, Flask, Django, model deployment.
- **Embedded & IoT:** Raspberry Pi, Arduino, STM32 (FreeRTOS, CAN Bus).

SYNERGISTIC ACTIVITIES

Nov 2026	Mentor, AI Hackathon, Heartland AI Symposium, Kansas State University (guiding teams on project scoping and technical questions)
May 2026	Vice-President, Badminton Club, Kansas State University
Oct 2025	Secretary, Computer Science Graduate Student Association (CS-GSA), Kansas State University
Feb 2023	Organizing Secretary, 2nd International Workshop on Computational Linguistics and Bangla Language Processing (CLBLP-2023)
Feb 2023	Organizing Member, 3rd International Conference on Electrical, Computer and Communication Engineering (ECCE-2023)
2022–2023	Vice-Chair (Activity), IEEE Young Professionals, Bangladesh Section
2019–2020	Branch Chairperson, IEEE CUET Student Branch
Feb 2019	Student Volunteer, International Conference on Electrical, Computer and Communication Engineering (ECCE 2019), Cox's Bazar, Bangladesh
2018–2019	Volunteer Instructor, robotics outreach for about 100 rural school students in Bangladesh: robotics concepts and hands-on work with Arduino, sensors, transistors and IR remote control
2016–Present	IEEE Student Member

ONLINE COURSES

Machine Learning in Production (DeepLearning.AI); Agentic AI (DeepLearning.AI); DeepLearning.AI TensorFlow Developer (Coursera); Machine Learning (Stanford University); Natural Language Processing in TensorFlow (Coursera); Convolutional Neural Networks in TensorFlow (Coursera); Data Scientist in Python (Dataquest.io).

STANDARDIZED TESTS

IELTS: Overall 8.0 (Listening 8.5, Reading 9.0, Writing 6.5, Speaking 7.0) – Test Report Form

REFERENCES

Dr. Pascal Hitzler (Ph.D. Advisor)
University Distinguished Professor and Lloyd T. Smith Creativity in Engineering Chair, Department of Computer Science; Director, Center for Artificial Intelligence and Data Science (CAIDS), Kansas State University, Manhattan, KS, USA
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Dr. Abhilekha Dalal
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